PRASITH BASKY MURUGAPERUMAL

MASTERS IN COMPUTATIONAL MECHANICS - UNI STUTTGART

Master's student in Computational Mechanics (Uni Stuttgart) seeking an internship and Master's thesis opportunity in engineering software development. Eager to leverage strong proficiencies in C++, Fortran, and Python to design, optimize, and implement scalable numerical methods, parallel algorithms, and core computational solvers.

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TECHNICAL SKILLS

- Programming: C++, Fortran, Python, Bash, MATLAB
- Parallelism & HPC: SLURM Job Scheduling, Cluster Environments (Ehlers), Git, CI/CD Pipelines
- Solvers & CAE: OpenRadioss, LS-DYNA, LS-OPT, Siemens Madymo, Ansys
- CAD Frameworks: Siemens NX, CATIA, Creo

EDUCATION

M.Sc. in Computational Mechanics (COMMAS)

University of Stuttgart | Oct 2024 – Present

- Continuum Mechanics
- Finite Element Methods
- Computational Fluid Mechanics
- Simulation software Engineering

B.E. in Mechanical Engineering

College of Engineering, Guindy – Anna University | Aug 2018 – Jul 2022

- GPA: 9.04/10 [1.58 German Grading]
- Gold Medalist – First Rank Holder
- Final Year Project: HVAC Design and Validation of College Auditorium using Ansys
- Project: CFD Analysis of Vortex Tube Refrigeration

LANGUAGES

- English – C1
- German – A2

PROFESSIONAL EXPERIENCE

WORKING STUDENT – ANSYS, PART OF SYNOPSYS)
AUG 2025 – PRESENT

- Configured and executed LS-DYNA finite element simulations to calibrate advanced constitutive material models, validating structural accuracy against complex experimental datasets.
- Automated high-dimensional parameter identification and optimization workflows using LS-OPT, applying numerical residual minimization algorithms to tightly correlate simulation response curves with material test metrics.
- Engineered a modular, object-oriented Python GUI application (Tkinter) to automate raw data ingestion, curve preprocessing, and report generation, transforming manual material calibration tasks into an integrated software pipeline.

STUDENT RESEARCH ASSISTANT
INSTITUTE OF TECHNISCHE UND NUMERISCHE MECHANIK (ITM)
NOVEMBER 2024 - JULY 2026

- Developed custom Python and Bash automation scripts for pre- and post-processing, reducing manual runtime configuration efforts for large-scale batch simulations
- Developed data validation pipelines to systematically compare extensive simulation measurement curves against experimental datasets for model refinement.
- Orchestrated high-fidelity LS-DYNA (THUMS) and Siemens Madymo simulations on the university's 'Ehlers' HPC cluster, managing parallel execution queues using SLURM batch scripts to evaluate structural crashworthiness and impact repeatability.

ASSOCIATE ENGINEER
CATERPILLAR INDIA PVT LTD
AUGUST 2022 - AUGUST 2024

- Conducted thermal, structural, and vibration analysis of engine components.
- Performed transient thermal analysis and temperature correlation with test data.
- Supported fatigue life prediction through thermo-mechanical analysis.
- Developed Python automation tools for CAE preprocessing and post-processing workflows.

RELEVANT PROJECTS & EXPERIENCE

Opensource Contributor OpenRadioss | Implicit solver development (https://github.com/prasithbasky/eigen_mat)

- Refactoring legacy numerical dependencies to decouple the solver path from the MUMPS library, increasing architectural modularity and introducing alternate high-performance sparse-matrix paths.
- Developing eigenvalue and eigenvector solver routines using LAPACK and ARPACK libraries in Fortran to optimize highly non-linear finite element computations.